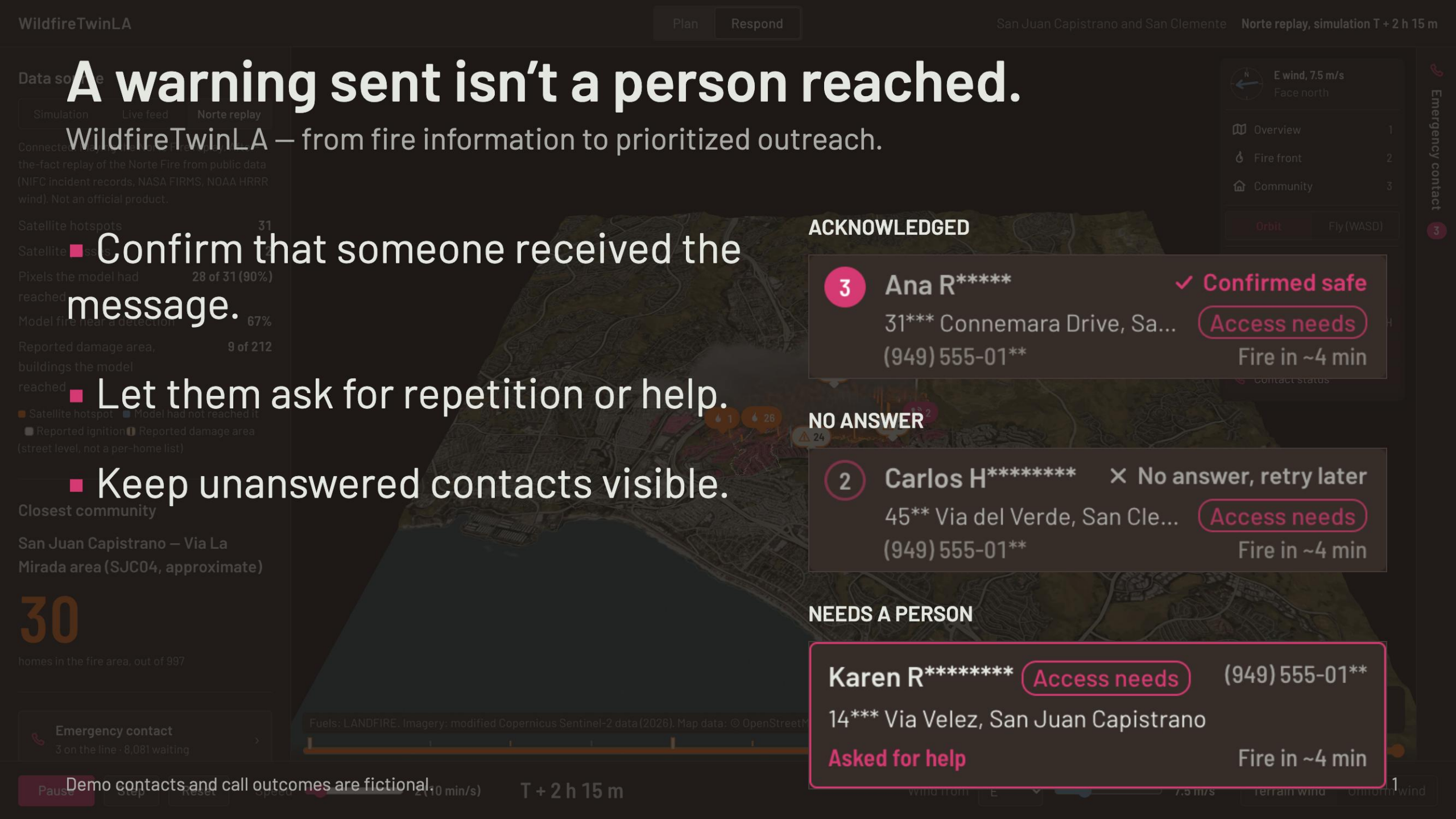
A warning sent isn't a person reached.

WildfireTwinLA — from fire information to prioritized outreach.

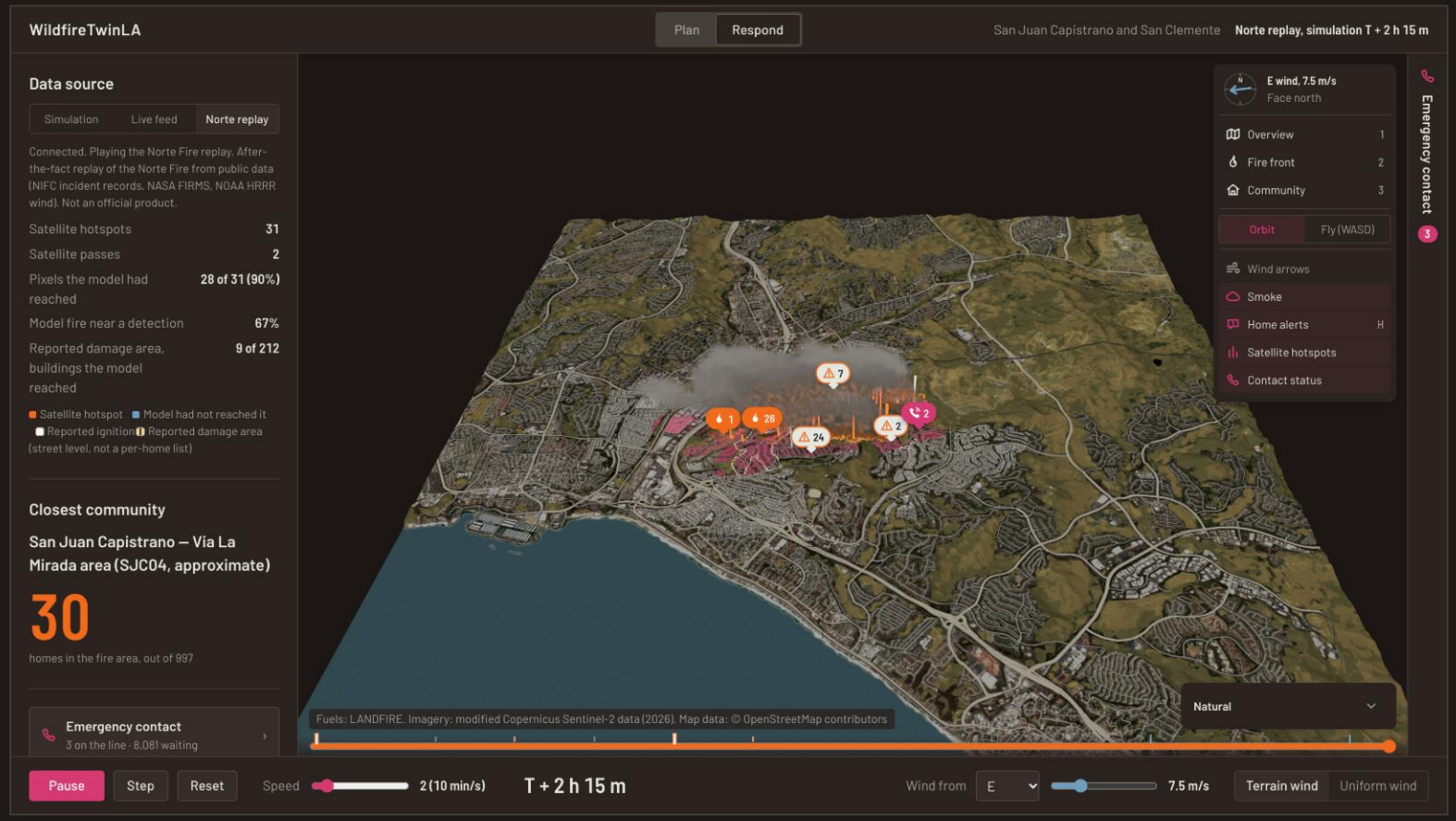
- Confirm that someone received the message.
- Let them ask for repetition or help.
- Keep unanswered contacts visible.



Demo contacts and call outcomes are fictional.

Real fires. A local decision.

Public fire observations, geographic context and experimental spread scenarios.



REPLAY

Norte Fire, Orange County — after-the-fact replay of Oct 16, 2026 at T + 2 h 15 m (≈1:48 p.m. PDT), from NIFC incident records, NASA FIRMS VIIRS and NOAA HRRR wind. Not live.

Regional context — CAL FIRE

Norte
San Clemente, Orange County · lead: OCFA
570 acres · 25% contained

Bryson
Malibu area, Los Angeles County
146 acres · 61% contained

Peak
Azusa area, Los Angeles County
43 acres · 41% contained

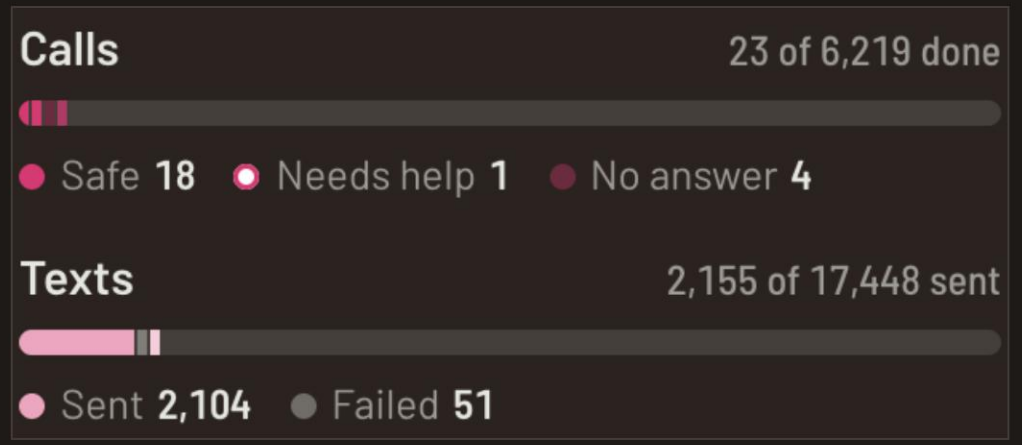
CAL FIRE pages (local agencies lead), updated Oct 17; retrieved 7:10–7:19 PM PDT. Bryson and Peak are not shown in our app.



Experimental, unofficial model outputs. Demo contacts and call outcomes are fictional. Follow official evacuation instructions.

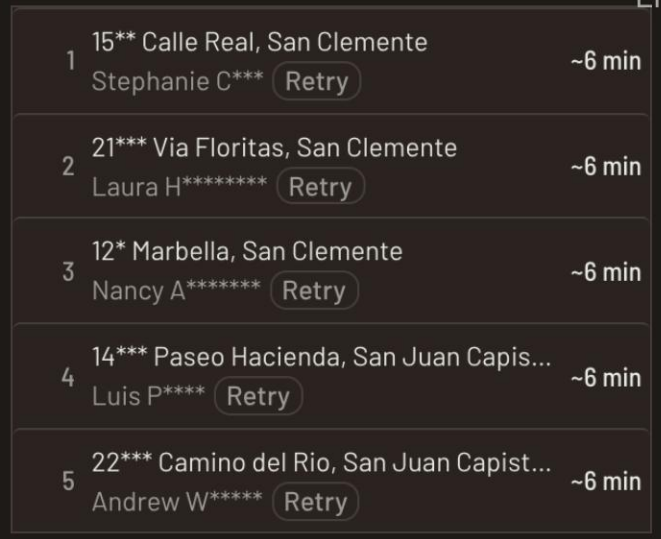
Limited staff. Many households. Who comes first?

- User: the overnight call-line supervisor.
- Decision: prioritize outbound follow-up.
- Delivery: simulated today; Retell integration planned.



Live counts from the simulated dispatcher





Ordered by estimated fire arrival

Retell voice call
Approved update; outcome returned to the queue

PLANNED Calls are simulated today

Ranking reflects estimated fire arrival, not conversational urgency or rescue priority. Demo contacts are fictional.

Nearby homes become a clear outreach queue.

Home location → Distance + wind (or model ETA) → Estimated arrival → Sorted queue → Call status



Each home's status sits above its roof (ring = queued).

1	15** Calle Real, San Clemente Stephanie C*** Retry	~6 min
2	21*** Via Floritas, San Clemente Laura H******** Retry	~6 min
3	12* Marbella, San Clemente Nancy A******* Retry	~6 min
4	14*** Paseo Hacienda, San Juan Capis... Luis P**** Retry	~6 min
5	22*** Camino del Rio, San Juan Capist... Andrew W***** Retry	~6 min

Rank · masked address · resident · flags · arrival

Top of the queue
earliest expected fire arrival

"Retry"
no answer on the first call

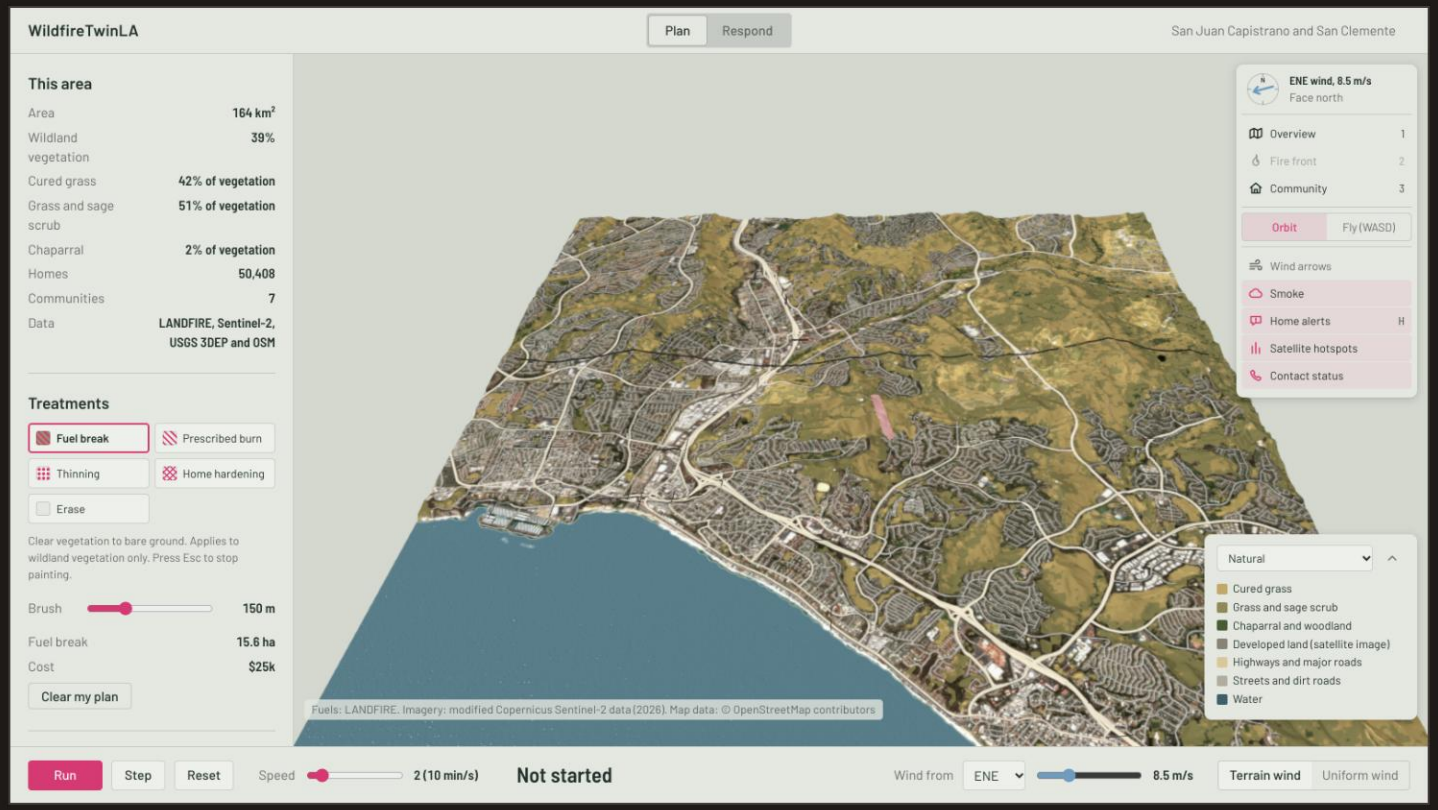
Farther away is not the same as safe.

Ranked by estimated fire arrival at each home (nearest simulated burning cell; simulation seeded from reported ignitions and VIIRS hotspots). Access-needs flag moves a home up 5 min. Not a complete danger assessment.

Experimental, unofficial model outputs. Demo contacts and call outcomes are fictional. Follow official evacuation instructions.

Before the emergency: where should we prepare?

- Intended user: community planner or Fire Safe Council.
- Explore preventive treatment locations.
- Compare scenarios before allocating resources.

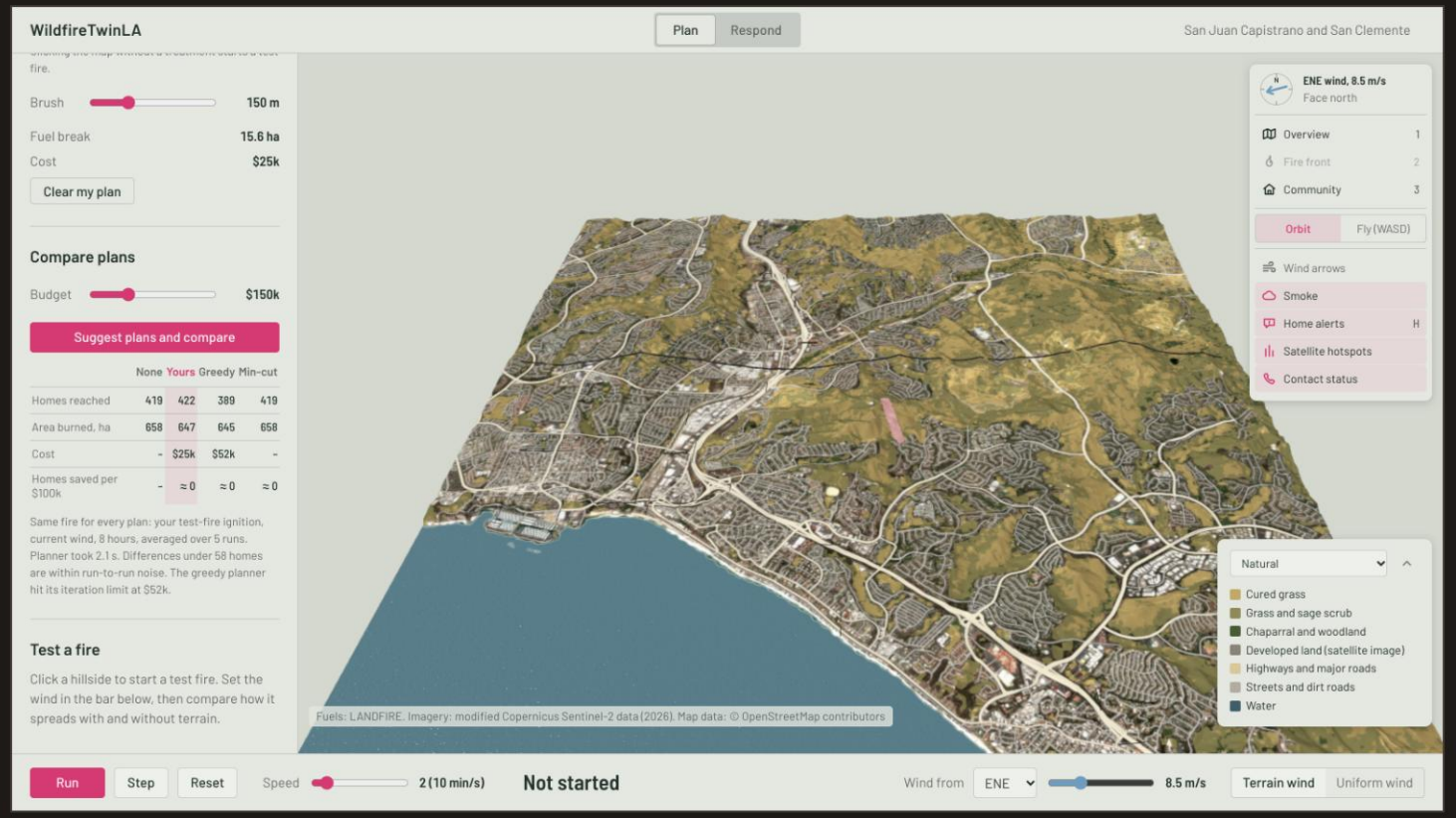


MODELLED Plan mode: a fuel break drawn downwind of the Norte ignition point

Prevention planning, not a guarantee. Prospective users are not confirmed partners.

Test a treatment. Compare the simulated outcome.

Fuel break · Prescribed burn · Thinning · Home hardening



	None	Yours	Greedy	Min-cut
Homes reached	419	422	389	419
Area burned, ha	658	647	645	658
Cost	-	$25k	$52k	-
Homes saved per $100k	-	≈ 0	≈ 0	≈ 0

Same fire for every plan: your test-fire ignition, current wind, 8 hours, averaged over 5 runs. Planner took 2.1 s. Differences under 58 homes are within run-to-run noise. The greedy planner hit its iteration limit at $52k.

Our $25k fuel break: 422 buildings reached vs 419 untreated — inside the ±58 run-to-run noise, so the app shows "≈ 0", not "homes saved".

MODELLED Test fire at the Norte ignition · ENE 8.5 m/s · 8 h · 5 runs · "homes" = mapped buildings

Experimental, unofficial model outputs. Demo contacts and call outcomes are fictional. Follow official evacuation instructions.

What powers today’s prototype?

Input	Function	Source / status
Terrain, fuels, buildings	Geographic context and home locations	USGS 3DEP, LANDFIRE FBFM40, OpenStreetMap (50,408 buildings), Sentinel-2 imagery (Oct 14, 2026) — committed real data
Satellite fire observations	Reported hotspot locations and times	NASA FIRMS VIIRS — replay of 4 passes on Oct 16; live FIRMS/GOES pollers built, not configured
Wind	Replay conditions and simulation input	Archived NOAA HRRR (5×5 points, hourly); WindNinja terrain-wind library
Resident contacts	Demonstrate the outreach workflow	Synthetic, one per building; fictional (949) 555-01xx numbers; never dialled

Observed data, model outputs and fictional contacts are labelled separately.

Community areas are approximate, not official evacuation zones. Replay evacuation orders: OCFA, as reported by CBS Los Angeles and KTLA (times approximate). Sentinel-2 is periodic imagery, not real-time monitoring.

What would an operational deployment need?

- Verified, authorized contact records.
- Agency-approved zones and message sources.
- Additional observations and independent validation.



Twist 1 — Federal funding pathway

Primary: USDA Forest Service Community Wildfire Defense Grant (CWDG). It funds Community Wildfire Protection Plans (up to $250k) and their projects; we would join an eligible community or Fire Safe Council application as the planning-and-outreach tool.

Alternative: NSF SBIR/STTR (America's Seed Fund), reauthorized through Sept 30, 2031; project pitches accepted anytime.

fs.usda.gov CWDG (next round not yet posted); sba.gov, Apr 13, 2026; seedfund.nsf.gov. Retrieved Oct 17, 2026.

Twist 2 — Going global (e.g., China)

Carries over: NASA FIRMS fire detections, Sentinel-2 imagery, Copernicus GLO-30 elevation; ECMWF or GFS weather replaces US-only HRRR.

Needs local work: fuel maps (LANDFIRE is US-only), building footprints (OSM is sparse in many regions), agency zones, contact consent, local telephony rules.

Retell voice agents support Mandarin (zh-CN).

NASA FIRMS; Copernicus; LANDFIRE; NOAA NCO; Herfort et al., Nature Comms 2023; docs.retellai.com. Retrieved Oct 17, 2026.

No agency relationship or funding award is implied. Building footprints do not establish a resident's phone number or permission to call.

No answer does not mean safe.

Call attempted

↓

No answer → retry once after 3 min — SIMULATED CONTACT

↓

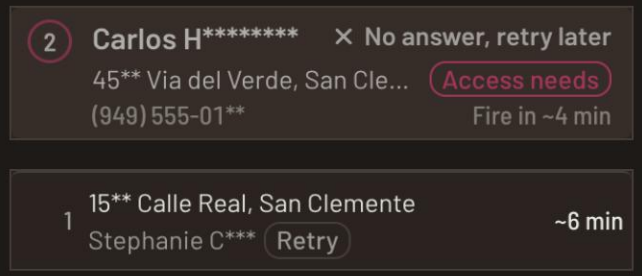
Still no answer → text moved up the queue — SIMULATED CONTACT

↓

Text delivered ≠ acknowledged

↓

Reply captured? Yes → record · No → unresolved — PLANNED



Line card and queue row in the demo: the retry stays visible to the operator.

Stale fire data → show age
PLANNED

Missing observations → simulation remains an estimate

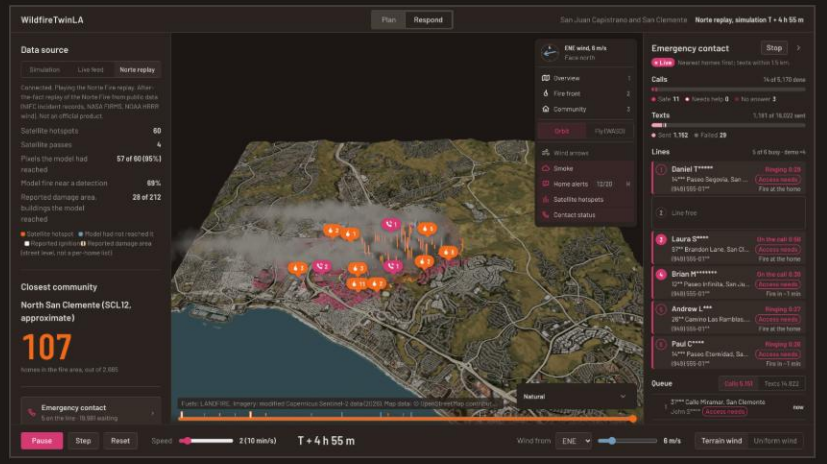
Phone/SMS outage → delivery unavailable

Experimental, unofficial model outputs. Demo contacts and call outcomes are fictional. Follow official evacuation instructions.

One workflow from preparation to outreach.

The map feeds an inspectable next action.

Tool category	What our design connects
Incident maps	Geographic context → household outreach queue
Notification systems	Recipient ordering → visible unresolved contacts
Prevention models	Treatment exploration → the same incident-area workflow



Prepare before the fire. Prioritize outreach during it.

Next: a consenting-volunteer pilot and verified call/SMS integration.

Team: Qinchuan · Tia · Lucia — The Dual-Use Hackathon, Resilience Comms lens

Code repository
github.com/ZQC8848/WildfireTwinLA

Category comparison, not a claim about named products. Fictional demo contacts; unofficial model outputs. Follow official evacuation instructions.